Rational equations

1 When is a rational expression undefined?

2 For each of the following, state the values which make the rational equations undefined:

a $\dfrac{6w}{w+4} = \dfrac{4}{w}$

b $-\dfrac{5k}{k+5} = \dfrac{2}{10k+3}$

c $\dfrac{2z}{z-3} = -\dfrac{1}{z}$

d $\dfrac{1}{3v} + \dfrac{10}{7} = -\dfrac{5}{v}$

e $m - \dfrac{6}{m} = 3$

f $\dfrac{b-2}{7b-2} + \dfrac{b-8}{6b} = -8$

g $\dfrac{6}{b-2} - \dfrac{7b}{(b-5)(b-8)} = 0$

h $\dfrac{p+4}{(p+4)(p+6)(p-9)} = 0$

i $-\dfrac{4}{j} + \dfrac{7}{j+5} = \dfrac{24}{j^2+j}$

j $\dfrac{4}{a+1} + \dfrac{a}{2+a} = \dfrac{a^2+49}{a^2-49}$

k $\dfrac{8}{q+3} - \dfrac{5}{q} = \dfrac{2}{q^2-9q}$

l $\dfrac{5}{q+3} + \dfrac{2}{q-9} = \dfrac{1}{q^2-3q-10}$

m $\dfrac{q^2-3q+8}{q(q^2-11q+28)} = 0$

n $\dfrac{5}{x-6} + \dfrac{9}{x+1} = \dfrac{7}{x^2+4x-45}$

3 Consider the equation $\dfrac{2}{5} + 4 = \dfrac{2}{x}$.
Determine whether the following could be the next step in solving for x:

a $\dfrac{2+20}{5} = \dfrac{2}{x}$

b $\dfrac{2x}{5x} + 4 = \dfrac{10}{5x}$

c $\dfrac{2}{5} - \dfrac{2}{x} = 4$

d $2x + 4 = 10$

4 Solve the following equations:

a $-\dfrac{7}{100} + \dfrac{x}{100} = \dfrac{7x}{100} + \dfrac{7}{100}$

b $\dfrac{1}{8}(2-6x) = \dfrac{1}{3}(8+5x)$

c $\dfrac{4n}{3} + \dfrac{1}{2} = \dfrac{6n-5}{6}$

d $\dfrac{x+2}{2} + 3 = \dfrac{x+3}{7}$

e $\dfrac{1}{n-2} = \dfrac{1}{9}$

f $\dfrac{7}{x+5} + \dfrac{3}{x+4} = 0$

g $\dfrac{1}{4m+4} + 2 = -\dfrac{1}{5}$

h $\dfrac{n-8}{n} + \dfrac{n+2}{n} = 5$

i $\dfrac{7}{n+1} - \dfrac{4}{n} = \dfrac{1}{n+1}$

j $\dfrac{n-1}{n} - \dfrac{n-2}{n+4} = \dfrac{1}{n}$

k $\dfrac{3x}{x-4} = \dfrac{1}{x+3} + 3$

l $3 - \dfrac{2}{x+7} = \dfrac{2x-1}{x-2}$

5 Kate performed the following steps to solve $\frac{x}{2}+\frac{x}{3}=\frac{2x}{5}$:

1. $\frac{x}{2}+\frac{x}{3}=\frac{2x}{5}$
2. $\frac{2x}{5}=\frac{2x}{5}$
3. $10x=10x$

Determine which step is incorrect and explain the error.

6 Consider the equation $\frac{x-1}{x-1}=5$.

a Yvonne is trying to solve the equation. Her next step is $x-1=5\left(x-1\right)$. For what values of x is this a valid step of work?

b How many solutions does the equation $\frac{x-1}{x-1}=5$ have?

7 Solve $\frac{ax}{b}+\frac{mx}{n}=6$ for x, expressing the solution in terms of a, b, m, and n.

8 For each of the following equations:

i Solve for the value(s) of the variable.

ii Determine the valid solution(s) to the equation.

a $\frac{2z}{z+3}-\frac{6}{z-3}=\frac{2z^2+18}{z^2-9}$

b $\frac{y+3}{y-3}-\frac{2}{y+3}-\frac{12}{\left(y+3\right)\left(y-3\right)}=\frac{5}{7}$

c $\frac{z-5}{z^2+9z+18}=\frac{1}{z+3}-\frac{1}{z+6}$

d $\frac{x+4}{x^2-12x+35}=\frac{1}{x-5}-\frac{1}{x-7}$

9 Find the solution set of the equation $\frac{4}{x^2-2x}+\frac{1}{x}=\frac{2}{x-2}$.

Applications

10 A construction company has spent $\$22\,500\,000$ to develop new cranes, and wants to limit the cost of development and production of each crane to $\$6000$.
Given that the production cost of each crane is $\$3000$, the cost for development and production of x cranes is given by $3000x + 22\,500\,000$ dollars, and so the cost of each crane is

$$\frac{3000x + 22\,500\,000}{x}$$

which we want to equal $\$6000$. Find the number of cranes that must be sold.

11 A driveway measuring $140\ \mathrm{ft}^2$ is to be paved. Harry has an apprentice working for him and divides the job between the two of them so that the area he paves divided by the area his apprentice paves equals 3. Find x, the area Harry will pave.

12 Beth and Elizabeth both played a season of basketball and finished the season having scored 32 points each. Beth played 8 games during the season, while Elizabeth played fewer games and had a game average that was 4 points greater than Beth's.
Find n, the difference in the number of games they played.